



UPM-ETSIAE

Systems Requirements Document

BlueHorizon 2024



Autores: ORTIZ TELLO, ALEJANDRO
VENERUSO, LUCA
MORENO GÓMEZ, JORGE
ARAMBURU CORBACHO, IKER
RUIZ SÁNCHEZ, JAVIER
PEDRAZA ABRIL, JOSÉ
FERNÁNDEZ LOJA, NOELIA

Chapter 1

Changelog

Author	Issue	Date	Pages	Change Description
Team 1	1.0	24/10/2024	Throughout	The following requirements have been modified: R-COM-10/1.1 R-COM-20/1.1 R-STR-10/1.1 R-STR-20/1.1 R-PWR-10/1.1 R-PWR-20/1.1 R-PRO-30/1.1 R-THR-10/1.1 R-AT-10/1.1 R-AT-20/1.1

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Chapter 2

Introduction

The Atlantic Ocean serves as a critical transportation route for passenger and goods between America, Europe and Africa. Recently, intelligent shipping has been introduced to maritime transport to monitor the condition of the containers with intelligent sensors allowing tracking important data like temperature, humidity, fire risks, among others.

However, communication with the containers is only possible within a few kilometres from the coast, leaving large gaps in data transmission for ships in the open ocean.

Therefore, tracking containers by satellite is of great interest. From this context, the Blue Horizon Mission emerged. The design of a satellite constellation to provide maritime communication coverage, using LoRa, to intelligent containers will support the demand for these devices in remote maritime regions allowing tracking and monitoring the devices continuously, ensuring the transmission of the data from any location along the route in the Atlantic Ocean, at any time of the day.



Chapter 3

Applicable and Reference documents

3.1 Applicable

- Technical Requirements Specifaction: ECSS-E-ST-10-06C [1]

3.2 Reference

- ExoMars2016-System Requirements Document
- STE-QUEST-Mission Requirements Document [6]
- Marco Polo-R Mission Requirements Document [3]
- IEEE (1984). IEEE Standard Letter Designations for Radar-Frequency Bands [2]
- NANOSTAR'S Project- Preliminary Design [5]
- State-of-the-Art of Small Spacecraft Technology NASA [4]

Chapter 4

General description

4.1 Scope of the document

This document is a Systems Requirements Document (SRD) to be used in the Blue Horizon Mission. The purpose of the SRD is to provide all high-level mission-level requirements, ensuring the design of the systems aligns with the mission objectives.

This document will serve as the basis for subsequent design phases and ensure all stakeholders maintain a shared understanding of the mission's objectives and constraints.

4.2 Mission Overview

Blue Horizon Mission is composed of a constellation of satellites in LEO orbit which aim to track the devices across the Atlantic for 10 years, providing continuous coverage along the maritime routes across this ocean. The main objectives of this mission are to:

- Receive and forward 10KB of data per device from 500 devices every hour.
- Communicate with LoRa devices.

Chapter 5

Specific requirements

5.1 Naming convention

The requirements will be identified with the following name: **SR-XXX-YY/VER**

Code	Name
XXX	Category of the requirement
YY	Enumeration of the requirement
VER	Version of the SRD in wich the requirement was introduced or lastly modified

Table 5.1: Naming convention of the requirements

5.2 Mission Analysis and Design requirements

R-MIS-10/1.1: The constellation shall receive and forward 10 KB of data per device from 500 devices in a single pass over the Atlantic Ocean.

This will limit the satellite viewing time.

R-MIS-20/1.1: The constellation shall be composed of less than 30 satellites.

R-MIS-30/1.1: The altitude of the satellites shall be less than 850km.

More than 850km of altitude is not compatible with LoRa technology.

R-MIS-40/1.1: Each satellite of the constellation shall be capable of storing the data of the 500 devices in a single pass over the Atlantic Ocean.

R-MIS-50/1.1: The constellation shall provide coverage of the entire Atlantic Ocean within a maximum of 2 hours.

5.3 Communications and Data Handling requirements

R-COM-10/1.2: The telecommunication equipment shall uplink (ground stations and LoRa) and downlink (ground stations) data at a frequency comprised in S-band.

The new LoRa 2.4 GHz technology operates in the S-band, eliminating the need to use different frequency bands, reducing complexity in frequency management and improving communication consistency across different regions. Additionally, the use of S-band is recommended for its low atmospheric attenuation.

R-COM-20/1.2: Each satellite of the constellation shall be capable of storing the LoRa devices data collected during a single pass over the Atlantic Ocean, as well as onboard telemetry and telecommands received from ground stations until the next pass over the Atlantic Ocean.

5.4 Structures and Mechanisms requirements

R-STR-10/1.2: The satellites shall be microsatellites up to 25 kg of wet mass and 400 mm X 250 mm X 250 mm envelop.

This requirement has been derived from the constraint of the launch system. It is unambiguous because the launcher upper stage is defined forcing the satellite envelope its volume requirement. In addition, the payload mass requirement of the launcher is defined. Both magnitudes, volume and mass, are quantitative and testable magnitude making this requirement verifiable.

R-STR-20/1.2: The spacecraft structure shall support the mechanical loads encountered during every mission phase.

This requirement has been derived from the constraint of the launch environment since the maximum loads appear in this phase. It is unambiguous because the launcher natural frequencies are defined, forcing the natural frequencies of the satellite to be above them with a positive Margin of Safety. In addition, natural frequencies are quantitative magnitudes; therefore, this requirement is verifiable.

R-STR-30/1.1: The spacecraft shall be designed in a modular way in order to obtain simple interfaces with the payload.

This requirement stems from the commercial objective of employing the satellite for secondary tasks. The several payloads shall be interchangeable in order to adapt the satellite to the different commercial purposes.

R-STR-40/1.1: The satellite shall integrate and protect the subsystems onboard.

This requirement is based on the limited space onboard and the necessity of incorporating every subsystem required for the mission and the different payload options for the commercial purpose. The volume available is defined by the R-STR-10, in addition volume is a quantitative and testable magnitude making this requirement verifiable.

5.5 Power Supply requirements

R-PWR-10/1.2: The power subsystem shall supply, distribute and condition the power in the satellite during all modes and phases of the mission to enable its operation.

The power subsystem must cope with the power needs of the various spacecraft subsystems including the periods of eclipse and the safe mode.

R-PWR-20/1.2: The power subsystem shall storage energy to ensure uninterrupted power supply to all the satellite's systems.

Any spacecraft requires a system to store energy for peak power demands and eclipse periods. This system must be recharged by the main power source when it is operational.

5.6 Propulsion requirements

R-PRO-10/1.1: The propulsion subsystem shall provide thrust to carry the satellites from the launcher injection orbit to their operating orbit.

The launcher carries the constellation of satellites from the ground to a LEO injection orbit. The propulsion subsystem of satellites provides thrust to perform the necessary maneuvers to reach the operating orbits.

R-PRO-20/1.1: The propulsion subsystem shall provide thrust to perform orbital corrections to maintain the operational orbit.

Due to the aerodynamic drag effects of the particles present in the operating orbits and other disturbance effects, the satellites may deviate from their established orbit. Therefore, the propulsion subsystem must provide thrust to correct these disturbances and keep the satellites in their operating orbit.

R-PRO-30/1.2: The propulsion subsystem shall provide thrust to deorbit the entire constellation from the end of its operation for up to 25 years of orbit.

Current legislation requires satellites to be removed from LEO orbits after a maximum of 25 years of operation. Hence, the propulsion subsystem must be designed to enable this final maneuver.

5.7 Thermal control requirements

R-THE-10/1.2: The spacecraft thermal control system shall ensure that the temperature of all spacecraft components remains within the specified operational ranges at all stages of the mission.

5.8 Attitude Determination Control requirements

R-AT-10/1.2: The attitude subsystem shall ensure the orientation of the antennas to guarantee full coverage.

R-AT-20/1.2: Ensure the orientation and starting of the propulsion subsystem for its operation.

Bibliography

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